HISTOLOGY PAST PAPER QUESTIONS COMPILED file 2;

—----------*by HMHDental*

Cell Biology & Microscopic Technique

Which of the following parts of a light microscope gathers light and creates a beam of light to focus it onto the specimen to be observed? (BDS Mid-term 2022)

a) Condenser

b) Diaphragm

c) Eyepiece

d) Illuminator

e) Objective

Correct Answer: a) Condenser

Reference: Laiq Hussain Histology, Chapter 1.

Explanation: The condenser is situated below the stage; its function is to collect and focus light from the source into a beam that illuminates the object.

Part of the microscope that converges light? (Anatomy mid 24)

a) Diaphragm

b) Condenser

c) Illuminator

d) Eyepiece

e) Objective

Correct Answer: b) Condenser

Reference: Laiq Hussain Histology, Chapter 1.

Explanation: Same principle as above; it converges light rays to provide a sharp image.

Which of the following is an acidic stain? (BDS Mid-term 2022)

a) Basic fuchsin

b) Eosin

c) Hematoxylin

d) Methylene blue

e) Toluidine blue

Correct Answer: b) Eosin

Reference: Laiq Hussain Histology, Chapter 1.

Explanation: Eosin is an anionic (acidic) dye that carries a negative charge and reacts with cationic groups (acidophilic components) like cytoplasmic proteins.

Which of the following is acidic dye? (Anatomy Mid 2025)

a) eosin

b) hematoxylin

c) methylene blue

d) basic fuchsin

e) toulidine blue

Correct Answer: a) eosin

Reference: Laiq Hussain Histology, Chapter 1.

Acidic dye used in slide preparation? (Anatomy mid 24)

a) Fuscin

b) Eosin

c) Hematoxylin

Correct Answer: b) Eosin

Reference: Laiq Hussain Histology, Chapter 1.

Which of the following is an acidophilic dye? (Anatomy mid 24)

a) Eosin

b) Hematoxylin

c) Methylene blue

Correct Answer: a) Eosin

Reference: Laiq Hussain Histology, Chapter 1.

Non-membrane bound organelle is: (Anatomy Mock 2024)

a) Centriole

b) Golgi body

c) Mitochondria

d) Ribosome

e) Rough endoplasmic reticulum

Correct Answer: d) Ribosome

Reference: Laiq Hussain Histology, Chapter 2.

Explanation: Ribosomes and centrioles lack a surrounding unit membrane. (Note: Centriole is also non-membrane bound, but Ribosome is the standard primary answer for this specific MCQ).

Which one of the following is non membrane bound organelles? (Anatomy Mid 2025)

a) mitochondria

b) SER

c) RER

d) ribosome

e) golgi bodies

Correct Answer: d) ribosome

Reference: Laiq Hussain Histology, Chapter 2.

Which of the following is a non-membrane bound organelle? (BDS Mid-term 2022)

a) Golgi apparatus

b) Mitochondrion

c) Ribosome

d) Rough endoplasmic reticulum

e) Smooth endoplasmic reticulum

Correct Answer: c) Ribosome

Reference: Laiq Hussain Histology, Chapter 2.

Cell organelle mainly involved in the synthesis of lipid is: (Anatomy Mid 2021)

a) Centrsome

b) Golgi apparatus

c) Mitochondria

d) Rough endoplasmic reticulum

e) Smooth endoplasmic reticulum

Correct Answer: e) Smooth endoplasmic reticulum

Reference: Laiq Hussain Histology, Chapter 2.

Explanation: SER contains enzymes for the synthesis of triglycerides, phospholipids, and steroid hormones.

Cell apoptosis (cell death) is carried out by: (Mock Dental Exam)

a) Peroxisomes

b) Smooth Endoplasmic Reticulum

c) Lysosomes

d) Golgi apparatus

e) Mitochondria

Correct Answer: c) Lysosomes

Reference: Laiq Hussain Histology, Chapter 2.

Explanation: Lysosomes contain hydrolytic enzymes (hydrolases) that break down cellular components during programmed cell death.

While observing cell under electron microscope the cytoplasm of cell appears basophilic, this property is due to: (Mock Dental Exam)

a) Smooth Endoplasmic Reticulum

b) Mitochondria

c) Rough Endoplasmic Reticulum

d) Lysosomes

e) Peroxisomes

Correct Answer: c) Rough Endoplasmic Reticulum

Reference: Laiq Hussain Histology, Chapter 2.

Explanation: Basophilia is due to the high RNA content of the ribosomes attached to the RER.

Example of microfilaments includes: (Anatomy Mid 2021)

a) Actin filaments

b) Desmin filaments

c) Keratin filaments

d) Neurofilaments

e) Vimentin filaments

Correct Answer: a) Actin filaments

Reference: Laiq Hussain Histology, Chapter 2.

Explanation: Microfilaments are the thinnest elements of the cytoskeleton, primarily composed of actin.

Centrioles made of: (Anatomy mid 24)

a) microtubules

Correct Answer: a) microtubules

Reference: Laiq Hussain Histology, Chapter 2.

Explanation: A centriole is a cylinder composed of nine triplets of microtubules.

DNA and histone (something): (Anatomy Mid 2019)

a) Nucleosome

Correct Answer: a) Nucleosome

Reference: Laiq Hussain Histology, Chapter 3.

Explanation: The fundamental unit of chromatin is the nucleosome, which consists of DNA wound around a core of histone proteins.

Mark the correct statement about different organelles of the cell: (Anatomy mock 2025-26)

a) Cristae are the infoldings of the outer membrane of mitochondria.

b) Centrioles are made up of microtubules.

c) Endoplasmic reticulum is studded with lysosomes on its surface.

d) Lysosomes are bounded by double membrane.

e) Nuclear envelope is connected to the golgi apparatus.

Correct Answer: b) Centrioles are made up of microtubules.

Reference: Laiq Hussain Histology, Chapter 2.

Epithelium & Glandular Histology

Pleura of lung is lined by: (Anatomy Mock 2024)

a) Pseudostratified columnar epithelium

b) Simple columnar epithelium

c) Simple cuboidal epithelium

d) Simple squamous epithelium

e) Transitional epithelium

Correct Answer: d) Simple squamous epithelium

Reference: Laiq Hussain Histology, Chapter 5.

Explanation: The simple squamous epithelium lining the serous cavities (pleura, pericardium, peritoneum) is called mesothelium.

Stratified columnar epithelium is found in: (Anatomy Mock 2024)

a) Alveoli of lung

b) Conjunctiva of eye

c) Esophagus

d) Follicles of thyroid gland

e) Uterus

Correct Answer: b) Conjunctiva of eye

Reference: Laiq Hussain Histology, Chapter 5.

Explanation: Stratified columnar is rare; it is found in the fornix of the conjunctiva and the large ducts of some glands.

Stratified columnar epithelium is found in the: (MOCK DDC 2025)

a) Alveoli of lung

b) Conjunctiva of eye

c) Distal convoluted tubules of kidney

d) Epidermis of skin

e) Uterus

Correct Answer: b) Conjunctiva of eye

Reference: Laiq Hussain Histology, Chapter 5.

Which type of epithelium lines the blood vessels? (Anatomy mock 2025-26)

a) Pseudostratified columnar epithelium

b) Simple columnar epithelium

c) Simple cuboidal epithelium

d) Simple squamous epithelium

e) Transitional epithelium

Correct Answer: d) Simple squamous epithelium

Reference: Laiq Hussain Histology, Chapter 5.

Explanation: This specific simple squamous epithelium is known as endothelium.

Simple cuboidal epithelium is found in: (Anatomy Mid 2021)

a) Conjunctiva

b) Epidermis

c) Esophagus

d) Gall bladder

e) Thyroid follicle

Correct Answer: e) Thyroid follicle

Reference: Laiq Hussain Histology, Chapter 5.

A researcher observes H&E stained tissue section of thyroid follicle in the light microscope. The lining epithelium most likely to be seen is: (Anatomy Mid 2025)

a) Pseudostratified columnar

b) Simple cuboidal

c) Simple columnar

d) Simple squamous

e) Transitional

Correct Answer: b) Simple cuboidal

Reference: Laiq Hussain Histology, Chapter 5.

Most appropriate histological feature of thyroid gland is the: (Anatomy mock 2025-26)

a) Absence of ducts.

b) Follicles filled with colloid.

c) Follicles lined by simple cuboidal epithelium.

d) Presence of parafollicular cells.

e) Richly vascular stroma.

Correct Answer: b) Follicles filled with colloid.

Reference: Laiq Hussain Histology, Chapter 21 (Endocrine).

Explanation: While c is also true, the "most appropriate" or "diagnostic" feature of the thyroid histology is the presence of follicles containing colloid.

Site of simple cuboidal? (Anatomy Mid 2019)

a) thyroid glad

Correct Answer: a) thyroid gland

Reference: Laiq Hussain Histology, Chapter 5.

Kidney is lined by? (Anatomy Mid 2019)

a) simple cuboidal

Correct Answer: a) simple cuboidal (referring to the tubules).

Reference: Laiq Hussain Histology, Chapter 5.

Simple columnar epithelium is found in: (Anatomy Mid 2025)

a) gall bladder

b) conjunctiva of eye

c) epidermis

Correct Answer: a) gall bladder

Reference: Laiq Hussain Histology, Chapter 5.

A researcher observes H&E stained tissue section of gall bladder under light microscope. The lining epithelium most likely to be seen is: (Anatomy Mid 2021)

a) Pseudostratified columnar

b) Simple cuboidal

c) Simple columnar

d) Simple squamous

e) Transitional

Correct Answer: c) Simple columnar

Reference: Laiq Hussain Histology, Chapter 5.

Epithelial lining of gallbladder? (Anatomy Mid 2019)

a) simple columnar

Correct Answer: a) simple columnar

Reference: Laiq Hussain Histology, Chapter 5.

Stomach is lined by: (Anatomy Mid 2021)

a) Pseudostratified columnar epithelium

b) Simple columnar epithelium

c) Simple cuboidal epithelium

d) Simple squamous epithelium

e) Transtional epithelium

Correct Answer: b) Simple columnar epithelium

Reference: Laiq Hussain Histology, Chapter 5.

Stratified squamous non-keratinized epithelium is found in: (BDS Mid-term 2022)

a) Conjunctiva

b) Epidermis

c) Esophagus

d) Gall bladder

e) Thyroid follicle

Correct Answer: c) Esophagus

Reference: Laiq Hussain Histology, Chapter 5.

A researcher observes H&E stained tissue section of urinary bladder under the light microscope. The lining epithelium most likely to be seen is: (BDS Mid-term 2022)

a) Pseudostratified columnar

b) Simple cuboidal

c) Simple columnar

d) Simple squamous

e) Transitional

Correct Answer: e) Transitional

Reference: Laiq Hussain Histology, Chapter 5.

Basal layer of epithelium is attached to the basement membrane with the help of which junction? (Anatomy Mid 2025)

a) gap junction

b) hemidesmosome

c) macula adherens

d) zonula adherens

e) macula

Correct Answer: b) hemidesmosome

Reference: Laiq Hussain Histology, Chapter 5.

Explanation: Hemidesmosomes resemble half-desmosomes and serve to anchor the basal cell membrane to the underlying basal lamina.

The basal surface of an epithelial cell is attached to the underlying basement membrane by a junction called: (Anatomy Mid 2021)

a) Gap junction

b) Hemidesmosome

c) Macula adherens

d) Zonula adherens

e) Zonula occludens

Correct Answer: b) Hemidesmosome

Reference: Laiq Hussain Histology, Chapter 5.

Which junction joins the epithelium with the basement membrane? (Anatomy mid 24)

a) Hemidesmosomes

Correct Answer: a) Hemidesmosomes

Reference: Laiq Hussain Histology, Chapter 5.

Submandibular salivary gland is a: (Anatomy Mock 2024)

a) Branched simple tubular gland

b) Coiled simple tubular gland

c) Compound acinar gland

d) Compound tubular gland

e) Compound tubuloacinar gland

Correct Answer: e) Compound tubuloacinar gland

Reference: Laiq Hussain Histology, Chapter 6.

Merocrine mode of secretion: (Anatomy Mid 2025)

a) sebaceous glands

b) salivary glands

c) sub mandibular glands

Correct Answer: b) salivary glands

Reference: Laiq Hussain Histology, Chapter 6.

Explanation: In merocrine secretion, the secretory product is released by exocytosis with no loss of cell substance.

Holocrine mode of secretion? (Anatomy mid 24)

a) sebaceous gland

Correct Answer: a) sebaceous gland

Reference: Laiq Hussain Histology, Chapter 6.

Explanation: In holocrine secretion, the entire cell disintegrates to release the secretory product.

Holocrine mode of secretion is found in which of the following glands? (MOCK DDC 2025)

a) Ceruminous glands of ear

b) Intestinal glands

c) Lacrimal glands

d) Mammary glands

e) Sebaceous glands

Correct Answer: e) Sebaceous glands

Reference: Laiq Hussain Histology, Chapter 6.

Apocrine mode of secretion is found in: (Anatomy Mid 2021)

a) Ceruminous glands of ear

b) Intestinal glands

c) Lacrimal glands

d) Mammary glands

e) Sebaceous glands

Correct Answer: d) Mammary glands

Reference: Laiq Hussain Histology, Chapter 6.

Explanation: In apocrine secretion, the apical portion of the cell is discharged along with the secretory product.

Uterine gland is the example of? (Anatomy Mid 2019)

a) simple tubular gland

Correct Answer: a) simple tubular gland

Reference: Laiq Hussain Histology, Chapter 6.

Connective Tissue & Cartilage

Tendon of a muscle is an example of: (Anatomy Mock 2024)

a) Adipose tissue

b) Loose connective tissue

c) Dense irregular connective tissue

d) Dense regular connective tissue

e) Reticular connective tissue

Correct Answer: d) Dense regular connective tissue

Reference: Laiq Hussain Histology, Chapter 7.

Explanation: Tendons consist of thick, parallel bundles of collagen fibers packed closely together with rows of fibroblasts in between.

Tendon is example of: (Anatomy Mid 2025)

a) dense irregular connective tissue

b) loose connective tissue

c) dense regular connective tissue

Correct Answer: c) dense regular connective tissue

Reference: Laiq Hussain Histology, Chapter 7.

Superficial and deep fascia are examples of: (Anatomy Mid 2021)

a) Adipose tissue

b) Loose connective tissue

c) Dense irregular connective tissue

d) Dense regular connective tissue

e) Reticular connective tissue

Correct Answer: b) Loose connective tissue

Reference: Laiq Hussain Histology, Chapter 7; BD Chaurasia General Anatomy.

Which cells have signet like appearance? (Anatomy Mid 2025)

a) Adipocytes

b) Fibroblast

c) Myofibroblast

d) Histiocytes

Correct Answer: a) Adipocytes

Reference: Laiq Hussain Histology, Chapter 7.

Explanation: The large fat droplet pushes the nucleus and a thin rim of cytoplasm to the periphery, giving it a ring-like appearance.

Signet ring appearance is the characteristic feature of the: (Anatomy mock 2025-26)

a) Adipocyte

b) Fibroblast

c) Histiocyte

d) Mast cell

e) Plasma cell

Correct Answer: a) Adipocyte

Reference: Laiq Hussain Histology, Chapter 7.

Regarding connective tissue cells and their functions, choose the incorrect pair: (BDS Mid-term 2022)

a) Adipocyte----------Storage of fat

b) Fibroblast----------Synthesis of extracellular fibers

c) Macrophage----------Phagocytosis

d) Mast cell----------Wound closure

e) Plasma cell----------Production of antibody

Correct Answer: d) Mast cell----------Wound closure

Reference: Laiq Hussain Histology, Chapter 7.

Explanation: Mast cells are involved in hypersensitivity and inflammatory reactions (releasing histamine/heparin), not wound closure.

Regarding connective tissue cells and their functions, choose the correct pair: (Anatomy Mid 2021)

a) Adipocyte----------Production of heparin

b) Fibroblast----------Phagocytosis

c) Macrophage----------Synthesis of extracellular fibers

d) Mast cell----------Storage of fat

e) Plasma cell----------Production of antibody

Correct Answer: e) Plasma cell----------Production of antibody

Reference: Laiq Hussain Histology, Chapter 7.

At which location hyaline cartilage usually does not have a perichondrium? (BDS Mid-term 2022)

a) Articular cartilage

b) Costal cartilage

c) Epiphyseal cartilage

d) Laryngeal cartilage

e) Nasal cartilage

Correct Answer: a) Articular cartilage

Reference: Laiq Hussain Histology, Chapter 8.

Explanation: Articular surfaces of bones are smooth and require the cartilage to be exposed directly to the joint cavity for reduced friction; thus, a fibrous perichondrium is absent.

Perichondrium is an example of: (Anatomy Mid 2021)

a) Adipose tissue

b) Dense irregular connective tissue

c) Dense regular connective tissue

d) Loose connective tissue

e) Reticular connective tissue

Correct Answer: b) Dense irregular connective tissue

Reference: Laiq Hussain Histology, Chapter 8.

Hyaline cartilage: (Anatomy Mid 2025)

a) Mainly composed of Type I collagen

b) Highly elastic and yellow in color

c) Glass-like appearance and has perichondrium

d) Found only in articular surfaces

e) Contains abundant adipose tissue

Correct Answer: c) Glass-like appearance and has perichondrium

Reference: Laiq Hussain Histology, Chapter 8.

Regarding histology of fibrocartilage, choose the statement which is false: (MOCK DDC 2025)

a) Chondrocytes are arranged in the form of rows

b) Contains mainly type-I collagen

c) Extracellular matrix is generally acidophilic

d) Extracellular matrix reveals parallel running bundles of collagen fibers

e) Perichondrium is usually identifiable

Correct Answer: e) Perichondrium is usually identifiable

Reference: Laiq Hussain Histology, Chapter 8.

Explanation: Fibrocartilage is unique because it lack a perichondrium; it usually merges with adjacent dense connective tissue or hyaline cartilage.

Example of fibrocartilage? (Anatomy mid 24)

a) intervertevral disc

Correct Answer: a) intervertebral disc

Reference: Laiq Hussain Histology, Chapter 8.

Bone & Muscle Histology

Regarding compact bone, choose the correct statement from the following: (Anatomy Mid 2025)

a) Periostium covers the bone matrix internally i.e around marrow cavity

b) Sharpey's fibers bind the bone matrix to periosteum

c) Volkman canal is present in the center of osteon

d) Osteocytes lie in the Howship's lacunae

e) Interstitial lamellae are present just beneath the periosteum

Correct Answer: b) Sharpey's fibers bind the bone matrix to periosteum

Reference: Laiq Hussain Histology, Chapter 9.

Explanation: Sharpey's fibers are bundles of collagen fibers that extend from the periosteum into the bone matrix to anchor it.

The periosteal arteries supply which area/part of a long bone? (MOCK DDC 2025)

a) Epiphysis

b) Inner 2/3rd of cortex

c) Medullary cavity

d) Metaphysis

e) Outer 1/3rd of cortex

Correct Answer: e) Outer 1/3rd of cortex

Reference: BD Chaurasia General Anatomy, Chapter 2.

Explanation: The periosteal arteries enter through Volkmann's canals to supply the outer layer of the compact bone.

Serology report of a 40 years old man reveals raised levels of parathyroid hormone. Which of the following bone cells is likely to be activated: (Anatomy Mid 2021)

a) Fibroblast

b) Osteoblast

c) Osteoclast

d) Osteocyte

e) Osteoprogenitor cell

Correct Answer: c) Osteoclast

Reference: Laiq Hussain Histology, Chapter 9.

Explanation: PTH stimulates osteoclast activity to increase bone resorption and raise blood calcium levels.

Common histological feature between skeletal and cardiac muscle: (Anatomy Mid 2025)

a) cross striations

b) uni nucleated

c) diads

d) muscle fibres

Correct Answer: a) cross striations

Reference: Laiq Hussain Histology, Chapter 10.

Explanation: Both skeletal and cardiac muscles are classified as "striated" because their myofibrils show a precise arrangement of actin and myosin.

What is true about the cardiac muscle fibers? (Anatomy mock 2025-26)

a) Are cylindrical and branching.

b) Contain nuclei at the periphery.

c) Show absence of gap junctions.

d) Have regenerative ability.

e) Have triads.

Correct Answer: a) Are cylindrical and branching.

Reference: Laiq Hussain Histology, Chapter 10.

Explanation: Cardiac muscle is characterized by its branching pattern and intercalated discs.

Specialized & Vascular Histology

Which of the following organs has fenestrated capillaries? (Anatomy mock 2025-26)

a) Bone marrow

b) Kidney

c) Liver

d) Lung

e) Skeletal muscle

Correct Answer: b) Kidney

Reference: Laiq Hussain Histology, Chapter 11.

Explanation: Fenestrated capillaries have pores to allow for rapid exchange or filtration, such as in the renal glomerulus.

Which organ has sinusoidal capillaries? (Anatomy Mid 2025)

a) Kidney

b) Lung

c) Liver

d) Brain

Correct Answer: c) Liver

Reference: Laiq Hussain Histology, Chapter 11.

Keratohyalin granules present in what layer: (Anatomy Mid 2025)

a) Stratum basale

b) Stratum spinosum

c) Stratum granulosum

d) Stratum lucidum

e) Stratum corneum

Correct Answer: c) Stratum granulosum

Reference: Laiq Hussain Histology, Chapter 15.

Explanation: Stratum granulosum is named for the basophilic keratohyalin granules found within the cells.

Which of the following lymphoid organs is hemi-capsulated? (Anatomy mock 2025-26)

a) Bone marrow

b) Lymph node

c) Spleen

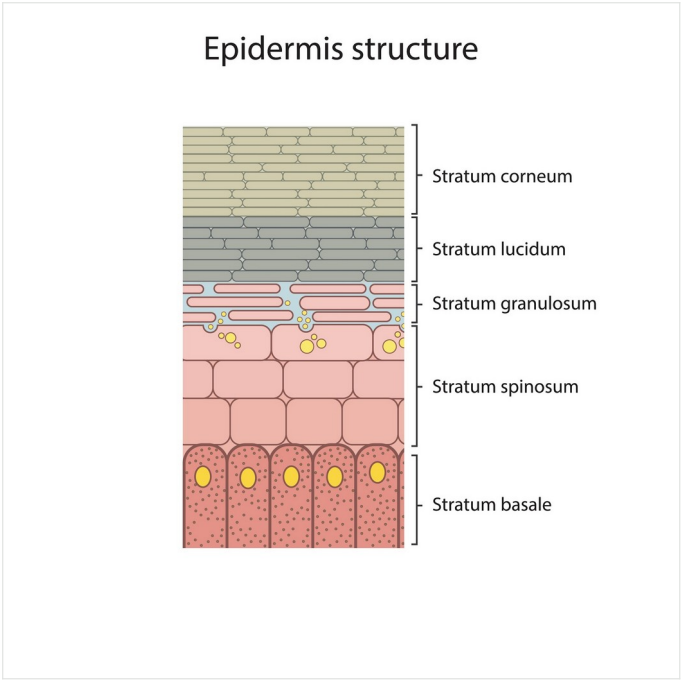
d) Thymus

e) Tonsil

Correct Answer: e) Tonsil

Reference: Laiq Hussain Histology, Chapter 13.

Explanation: The palatine tonsil is covered by epithelium on its free surface and a partial (hemi) connective tissue capsule on its deep surface.



FRIENDLY ADVICE:

Yeah too much to handle right??? If you made this far then i have a friendly advice for ya Focus on Epithelium examples and Glands examples for finals, believe me weightage km he Histo ki in anatomy paper, focus more on Head and Neck(35) and Embryo(15)